CS 839 · NEXT GENERATION SYSTEMS · MIDTERM · 5 NOVEMBER 2026

PokéSearch

Evolutionary search for “find all” queries

Finding every match without reading everything.

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01 · THE PROBLEM

Some questions need every match, not the best ten

Ordinary search

Ranks the top few. One good answer is enough.

“Find all” queries

Need every qualifying document. **Every miss fails silently.**

Two options today, and both are bad:

Search — cheap, and you never learn what it missed.

Read everything — complete, and **~$532** for these 3,000 filings with a frontier judge. **95.7%** of that effort lands on filings that turn out to be irrelevant.

Cost scales linearly in corpus size **and** in judge price — and those multiply.

01 · THE PROBLEM

Every retrieval method is blind somewhere

Exact phrase

`stay-at-home`

stay-at-home orders	match
stay at home	no match
staying home	no match
stayed at home	no match

Complete for one string — and blind to every other way of saying it.

Keyword · BM25

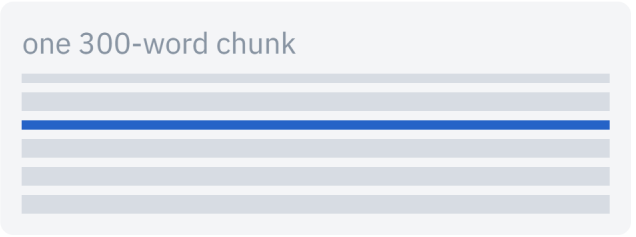
the query
`covid · hobby · lifestyle · increase`

no shared words

a filing that answers it
`bicycle · sales · surged · families · outdoor`

Ranks by word overlap. Same meaning, nothing in common, no result.

Embeddings

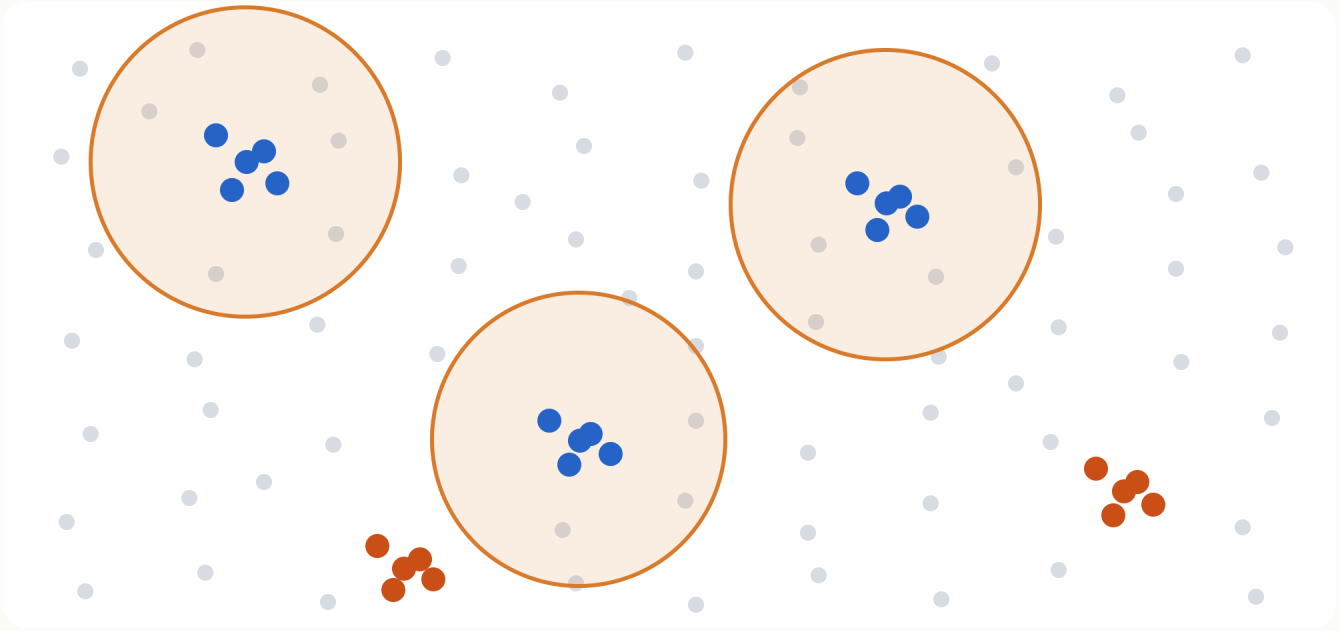


averaged into

One vector for 300 words. The one sentence that matters is diluted by the 290 that don’t.

01 · THE PROBLEM

And the answers are not in one place



Blue and **dark orange** are the filings that answer the question. They sit in several unrelated pockets — one per vocabulary.

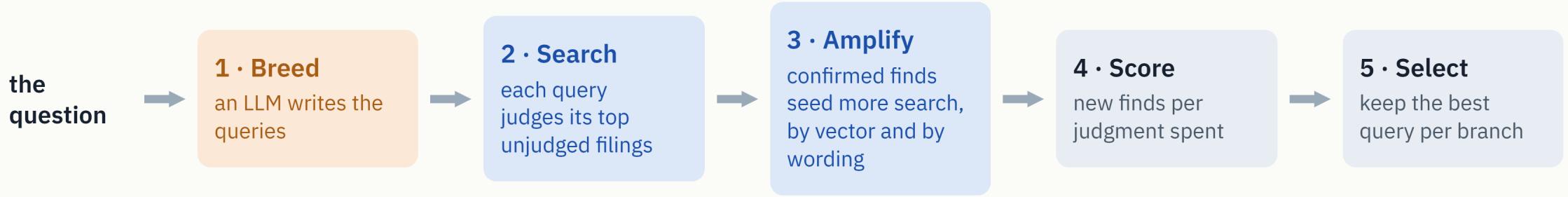
The circles are what our queries retrieve: each one reaches its own neighbourhood and nothing else.

The dark orange pockets are reached by nothing we wrote — not hidden, just never near any query.

We can’t change the space. We can move the query.

02 · OUR METHOD

Treat the query set as a population that evolves



↻ back to Breed, until the judgment budget runs out · **generation 0 is step 1 with nothing to learn from yet**

Diverse population

Queries spread across exact phrase, keyword and embedding search, so no method dies out.

Novelty fitness

A query is scored on filings **nobody has seen yet**. Re-finding what you already have scores zero.

Elite breeding

Winners survive unchanged and push deeper into their own rankings until they run dry.

`settings are derived, not chosen:  generations = budget ÷ (population × judge depth)  →  1,000 ÷ (6 × 25) ≈ 7`

02 · THE BENCHMARK

3,000 SEC filings, one question, one fixed budget

3,000
2020 10-K filings, Items 1A & 7

129
relevant filings (4.3%)

270,029
indexed 300-word chunks

1,000
judgments per run — 1 judgment = 1 filing read

`stay_home_boom`

Did consumer pandemic lifestyle changes drive higher purchases of the company's products?

We chose a corpus small enough to have a complete answer key. That is what lets us measure recall exactly instead of estimating it.

Fair by construction

Same six starting queries and the same budget for every arm.

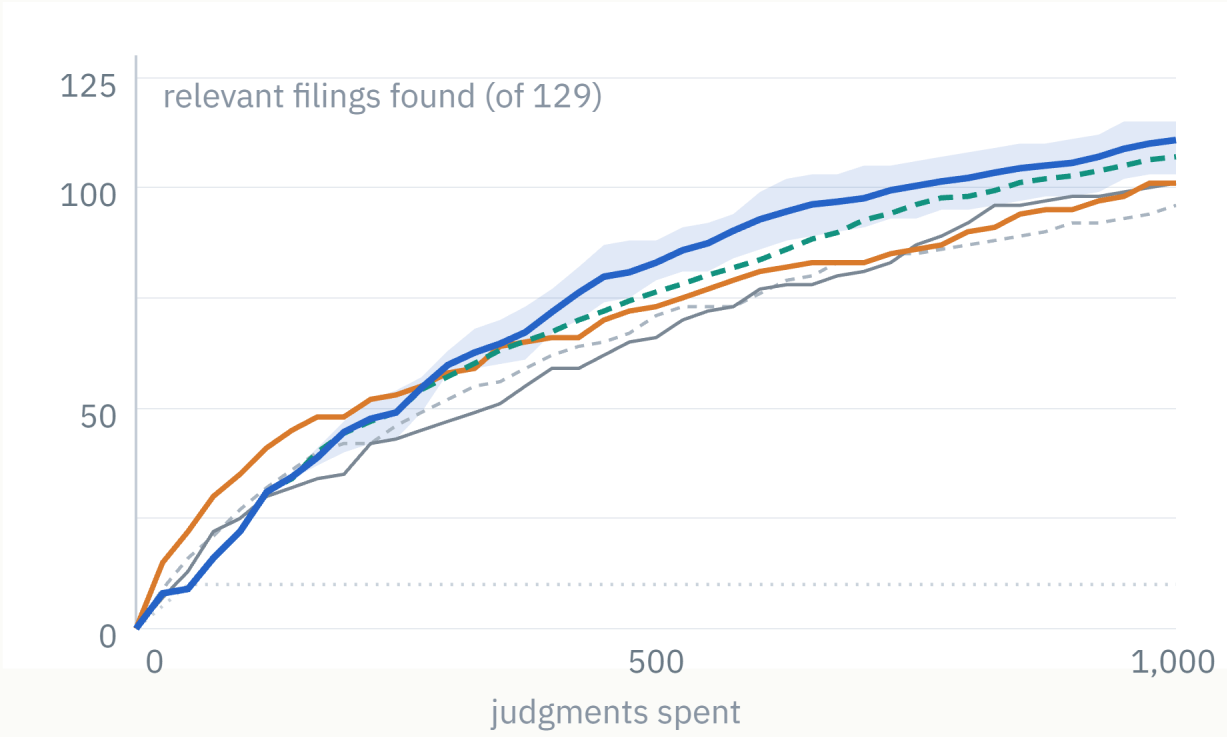
The breeder sees only queries and judged excerpts — never labels, never what remains.

The judge is a perfect lookup, so our recall is an upper bound.

Answer key written once by a cheap LLM over every filing; input hashes and commit SHA logged on every run.

03 · RESULTS

It beats every static baseline — and 9 of 10 runs beat them outright



At 1,000 judgments	Found	Recall
Exact phrase	10	7.8%
Embeddings	96	74.4%
BM25	101	78.3%
Hybrid, all three	101	78.3%
PokéSearch · Haiku	107.0	82.9%
PokéSearch · Sonnet	110.8	85.9%

Best single run: **115 of 129 — 89.1%**

Our worst run found 103. The best baseline found 101.

03 · RESULTS · ONE RUN, GENERATION BY GENERATION

Each generation opened a category we never wrote down

GEN 0	+31 → **31**	new	bm25	8	·	covid, hobby, lifestyle, increase
		new	embed	7	·	increased revenue due to COVID related consumer habits
GEN 1	+15 → **46**	new	bm25	5	·	consumers pandemic gardening baking cooking puzzles crafts pets
		kept	embed	5	·	increased revenue due to COVID related consumer habits
GEN 2	+22 → **68**	new	bm25	8	·	pandemic homebound consumers boating RVs bikes fitness equipment…
		new	embed	6	·	people spending more time at home… streamed more, played video games…
GEN 3	+21 → **89**	new	bm25	7	·	home improvement projects homeowners paint flooring furniture…
		kept	embed	5	·	people spending more time at home… streamed more, played video games…
GEN 4	+11 → **100**	new	bm25	4	·	swimming pools backyard hot tubs grills patio staycation demand…
		new	embed	3	·	consumers cooking and baking at home… grocery, pantry, meal kits…
GEN 5	+7 → **107**	new	embed	2	·	games, puzzles, crafts, toys, books and hobby products rose…
		kept	embed	2	·	people spending more time at home… streamed more, played video games…
GEN 6	+8 → **115**	new	embed	3	·	increased at-home consumption of food, beverages, snacks, groceries…
		kept	embed	2	·	people spending more time at home… streamed more, played video games…

The LLM walked into categories we never wrote down

56.5% of all finds came from queries **the LLM bred** — not the six we wrote.



GEN 0 · hand-written · 31 found
covid, hobby, lifestyle, increase
(Initial Covid-19 stay_home_boom)



PokéSearch Breeder
(Sonnet 5.5 — $0.12/run)



GEN 1 · 46 found
consumers pandemic gardening baking pets (+15 new)



GEN 3-4 · 100 found
swimming pools backyard hot tubs grills...
(Home improvement, grills & pools • +32 new)





GEN 2 · 68 found
(RVs, boating, bikes & streaming • +22 new)

GEN 5-6 · 115 found
increased at-home consumptiion of food...
(At-home snacks & workouts • 115/129 found!)

04 · LIMITS

What we can’t claim yet

Scale

Our premise is that reading everything is prohibitive. On 3,000 filings it costs $15.63. **We validated the mechanism where we could measure it, not where it pays off.**

Machinery vs. model

An LLM rewriting queries each round — no population, no fitness, no selection — might match us. **We’ve beaten static search, not LLM-driven search.**

A perfect judge

It never errs, and it answers for the whole filing. Recall is an upper bound.

Silver labels

One question, one corpus, and an answer key written by a single cheap model.

No stopping signal

Recall swings 103 to 115 run to run, and we can’t tell you which one you got.

04 · WHAT'S NEXT

What we build next

01

Iterative-rewriting baseline

Same seeds, same budget, same model — only the evolutionary scaffolding removed.

Why first: it is the one experiment that decides whether any of this machinery earns its keep.

02

A real LLM judge

Snippet first, full text when the snippet is not enough.

Why: every recall number we showed today is an upper bound until this exists.

03

Random immigrants

Sample regions no query has touched; relevant hits re-enter as fresh seeds.

Why: evolution can only move from where it has already been.

04

Scale and generality

More questions, a multi-hop query, and a corpus where reading everything is genuinely out of reach.

Why: that is the setting the method was designed for.

Thank you — questions?